

Matematika Sains

DATA

UJIAN
TENGAH
SEMESTER

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PARAMETER PERSONALISASI

- $N \text{ (NIM mod } 100)$

$$N = 19 \text{ mod } 100 = \underline{\underline{19}} \rightarrow 19 : 100 = 0 \text{ sisa } (19)$$

- $a \text{ (floor } (N/10))$

$$a = \lfloor 19/10 \rfloor = 1,9 \approx \underline{\underline{1}} \text{ (pembulatan ke bawah)}$$

- $b \text{ (N mod } 10)$

$$b = 19 \text{ mod } 10 = \underline{\underline{9}} \rightarrow 19 : 10 = 1 \text{ sisa } (9)$$

- $K \rightarrow (a+b+1)$

$$K = 1 + 9 + 1 = \underline{\underline{11}}$$

- $\theta \rightarrow (a \cdot 30) + (b \cdot 5)$

$$\theta = (1 \cdot 30) + (9 \cdot 5) = 30 + 45 = \underline{\underline{75^\circ}}$$

- $\alpha \rightarrow 0,001 \cdot (b+1)$

$$\alpha = 0,001 \cdot (9+1) = 0,001 \times 10 = \underline{\underline{0,01}}$$

② Parameter : $a = 1, b = 9, N = 19$

$$a) \quad A = \begin{bmatrix} a+1 & b & 2 \\ 1 & a+2 & b \\ b & 1 & a+3 \end{bmatrix} = \begin{bmatrix} 1+1 & 9 & 2 \\ 1 & 1+2 & 9 \\ 9 & 1 & 1+3 \end{bmatrix} = \begin{bmatrix} 2 & 9 & 2 \\ 1 & 3 & 9 \\ 9 & 1 & 4 \end{bmatrix}$$

$$\det(A) = 2 \begin{vmatrix} 3 & 9 \\ 1 & 4 \end{vmatrix} - 9 \begin{vmatrix} 1 & 9 \\ 9 & 4 \end{vmatrix} + 2 \begin{vmatrix} 1 & 3 \\ 9 & 1 \end{vmatrix} = 2(12-9) - 9(4-81) + 2(27-1)$$

$$= 2(3) - 9(-77) + 2(26) = 6 + 693 + 52 = 647 //$$

b) Minor	Matrices	Hitungan	Hasil
M_{11}	3, 9, 1, 4	$(3 \cdot 4) - (1 \cdot 9) = 12 - 9$	3
M_{12}	1, 9, 9, 4	$(1 \cdot 4) - (9 \cdot 9) = 4 - 81$	-77
M_{13}	1, 3, 9, 1	$(1 \cdot 1) - (3 \cdot 9) = 1 - 27$	-26
M_{21}	9, 2, 1, 4	$(9 \cdot 4) - (2 \cdot 1) = 36 - 2$	34
M_{22}	2, 2, 9, 4	$(2 \cdot 4) - (2 \cdot 9) = 8 - 18$	-10
M_{23}	2, 9, 9, 1	$(2 \cdot 1) - (9 \cdot 9) = 2 - 81$	-79
M_{31}	9, 2, 3, 9	$(9 \cdot 9) - (2 \cdot 3) = 81 - 6$	75
M_{32}	2, 2, 1, 9	$(2 \cdot 9) - (2 \cdot 1) = 18 - 2$	16
M_{33}	2, 9, 1, 3	$(2 \cdot 3) - (9 \cdot 1) = 6 - 9$	-3

c) • Matriks kofaktor C

$$\begin{bmatrix} +M_{11} & -M_{12} & +M_{13} \\ -M_{21} & +M_{22} & -M_{23} \\ +M_{31} & -M_{32} & +M_{33} \end{bmatrix} = \begin{bmatrix} 3 & 77 & -26 \\ -34 & -10 & 79 \\ 75 & -16 & -3 \end{bmatrix}$$

• Matriks adjoin $\text{adj}(A) = C^T$

$$\text{adj}(A) = \begin{bmatrix} 3 & -34 & 75 \\ 77 & -10 & -16 \\ -26 & 79 & -3 \end{bmatrix}$$

• Matriks invers

$$A^{-1} = \frac{1}{\det(A)} \cdot \text{adj}(A) = \frac{1}{647} \cdot \begin{bmatrix} 3 & -34 & 75 \\ 77 & -10 & -16 \\ -26 & 79 & -3 \end{bmatrix} = \begin{bmatrix} 3/647 & -34/647 & 75/647 \\ 77/647 & -10/647 & -16/647 \\ -26/647 & 79/647 & -3/647 \end{bmatrix} //$$

$$e) \quad P = \begin{bmatrix} 120 & 50+N & 200 \\ 80 & a+2 & 10+N \\ 30+N & 90 & 170 \end{bmatrix} = \begin{bmatrix} 120 & 50+19 & 200 \\ 80 & 1+2 & 10+19 \\ 30+19 & 90 & 170 \end{bmatrix} = \begin{bmatrix} 120 & 69 & 200 \\ 80 & 3 & 29 \\ 49 & 90 & 170 \end{bmatrix}$$

$$E = A \cdot P = \begin{bmatrix} 2 & 9 & 2 \\ 1 & 3 & 9 \\ 9 & 1 & 4 \end{bmatrix} \cdot \begin{bmatrix} 120 & 69 & 200 \\ 80 & 3 & 29 \\ 49 & 90 & 170 \end{bmatrix}$$

→
NEXT

$$e_{11} = (2.110) + (9.80) + (2.49) = 240 + 720 + 98 = 1058$$

$$e_{12} = (2.69) + (9.3) + (2.90) = 138 + 27 + 180 = 345$$

$$e_{13} = (2.200) + (9.29) + (2.170) = 400 + 261 + 340 = 1001$$

$$e_{21} = (1.170) + (3.80) + (9.49) = 120 + 240 + 441 = 801$$

$$e_{22} = (1.69) + (3.3) + (9.90) = 69 + 9 + 810 = 888$$

$$e_{23} = (1.200) + (3.29) + (9.170) = 200 + 87 + 1530 = 1817$$

$$e_{31} = (9.170) + (1.80) + (4.49) = 1080 + 80 + 196 = 1356$$

$$e_{32} = (9.69) + (1.3) + (4.90) = 621 + 3 + 360 = 984$$

$$e_{33} = (9.200) + (1.29) + (4.170) = 1800 + 29 + 680 = 2509$$

$$E = \begin{bmatrix} 1058 & 345 & 1001 \\ 801 & 888 & 1817 \\ 1356 & 984 & 2509 \end{bmatrix} //$$

• Deskripsi $P' = A^{-1} \cdot E$

$$P' = \begin{bmatrix} 3/647 & -34/647 & 75/647 \\ 76/647 & -10/647 & -16/647 \\ -26/647 & 79/647 & -3/647 \end{bmatrix} = \begin{bmatrix} 1058 & 345 & 1001 \\ 801 & 888 & 1817 \\ 1356 & 984 & 2509 \end{bmatrix}$$

membuktikan $P' = P$

$$P'_{11} = \left(\frac{3 \cdot 1058}{647} \right) + \left(\frac{-34 \cdot 801}{647} \right) + \left(\frac{75 \cdot 1356}{647} \right) = \frac{77640}{647} = 120 \text{ (sesuai)}$$

$$P'_{12} = \left(\frac{3 \cdot 345}{647} \right) + \left(\frac{-34 \cdot 888}{647} \right) + \left(\frac{75 \cdot 984}{647} \right) = \frac{44647}{647} = 69 \text{ (terbukti)}$$

$$P'_{13} = \left(\frac{3 \cdot 1001}{647} \right) + \left(\frac{-34 \cdot 1817}{647} \right) + \left(\frac{75 \cdot 2509}{647} \right) = \frac{129400}{647} = 200 \text{ (terbukti)}$$

$$P'_{21} = \left(\frac{76 \cdot 1058}{647} \right) + \left(\frac{-10 \cdot 801}{647} \right) + \left(\frac{-16 \cdot 1356}{647} \right) = \frac{51760}{647} = 80 \checkmark$$

$$P'_{22} = \left(\frac{76 \cdot 345}{647} \right) + \left(\frac{-10 \cdot 888}{647} \right) + \left(\frac{-16 \cdot 984}{647} \right) = \frac{1941}{647} = 3 \checkmark$$

$$P'_{23} = \left(\frac{76 \cdot 1001}{647} \right) + \left(\frac{-10 \cdot 1817}{647} \right) + \left(\frac{-16 \cdot 2509}{647} \right) = \frac{18763}{647} = 29 \checkmark$$

$$P'_{31} = \left(\frac{-26 \cdot 1058}{647} \right) + \left(\frac{79 \cdot 801}{647} \right) + \left(\frac{-3 \cdot 1356}{647} \right) = \frac{31703}{647} = 49 \checkmark$$

$$P'_{32} = \left(\frac{-26 \cdot 345}{647} \right) + \left(\frac{79 \cdot 888}{647} \right) + \left(\frac{-3 \cdot 984}{647} \right) = \frac{58230}{647} = 90 \checkmark$$

$$P'_{33} = \left(\frac{-26 \cdot 1001}{647} \right) + \left(\frac{79 \cdot 1817}{647} \right) + \left(\frac{-3 \cdot 2509}{647} \right) = \frac{109990}{647} = 170 \checkmark$$

$$P' = P = \begin{bmatrix} 120 & 69 & 200 \\ 80 & 3 & 29 \\ 49 & 90 & 170 \end{bmatrix} //$$

- Jika $\det(A) = 0$

Yang akan terjadi jika $\det(A) = 0$ adalah matriks A tidak akan memiliki invers sehingga proses deskripsi (p') tidak dapat dilakukan yang seharusnya dapat mengembalikan data ke bentuk aslinya ($p' = p$).